
PROFOUND NOVELTY · AMENDMENT A4D

The Alien Reading, Executed

The standing offer accepted: the kit's verbatim prompt
carried to a reader of another model family, the reading
returned through the owner's repository — three points
below the family band, zero inversions, three cells
moved. And the fifth pair admitted: caloric against
thermodynamics, the strongest corpse and its
executioner.

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FIVE PAIRS · TEN THEORIES · TWO LINEAGES · SIX READINGS

A STANDALONE ARTIFACT · SEPTEMBER 2026

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1. The Mandate: The Offer Was Accepted

The offer was accepted, and this document opens with the receipt. The Widened Census ended by converting its unfilled alien clause into a standing kit: the verbatim blind-coding prompt, the panel of eight, the scale, the evidence rule, the scoring scheme, and the adjudication rules printed and archived so that any genuinely external reader could execute the alien reading without the author in the loop. That is now what has happened. The series' owner carried the kit out of this environment, administered it to a reader of a different model family — the returned file's own first line names its reader gpt — and delivered the complete output back through the owner's repository. What arrived is exactly what the kit demanded: the nine-by-eight matrix, a named, checkable historical fact for every cell, and point notes that read the instrument with the eyes of a lineage this series has never run. The genetic-dependence limit — the census's loudest confession since A4b, a demonstrated availability fact since A4c — is broken as an availability fact. The harder test the series promised to face on that day has arrived, and this amendment faces it.

Any genuinely external reader — a human historian, or a model from another family, run by anyone, anywhere — can execute the alien reading without the author in the loop.

The Widened Census, Chapter 6 — the standing offer, accepted before this document was written; this amendment opens with the receipt and closes with the re-issue.

The second half of the order was named in the same breath: the next matched pair, caloric against thermodynamics, the natural candidate, per the census's own extension rule. The rule executes unchanged. Thermodynamics enters the case base by Volume II's method for admitting residents — ratified by the world rather than by this series: a century and a half of executed physics and engineering, generative without any degenerative phase, the executioner of a theory that qualifies as a control. Caloric enters as the fifth dead theory under the census's four sampling criteria. And the census re-runs under the same criteria, with the same coding manual, and prints whatever differs. What differs turns out to be more than the pair itself, because the fifth pair is the first whose dead side was fully quantitative — a corpse with theorems, instruments, and minted units of its own — and the census had never before had to say what it thinks survives when a quantitative theory dies.

What the alien reading can and cannot be certified to be is stated now, before anything is done with it, because provenance is this series' first discipline. The administration chain: the kit's prompt, archived verbatim by A4c, was carried by the series owner to a session outside this environment; the session's reader is identified by the returned file's own label, gpt; the output was committed to the owner's repository and pulled by this author. The blinding conditions were the kit's own — no files, no web, no lookups, every cell demanding named evidence — and the reading's closing line records an interface marker this environment's readers do not have (no Write tool available in that session), the signature of a

genuinely different runtime. The limits print with the same brush: the family label is attested by the delivery, not verifiable from inside this environment, and the census says so plainly. What the census does not do is decline the reading it spent an amendment arming itself to receive.

The order of business follows the series' discipline, and the pre-registration came first of all. The author's codings for caloric and thermodynamics, with the expected fault lines and their permitted resolutions, were locked to file before the fourth blind coding ran — with the contamination statement printed inside it: the author had already read the alien's eight-theory output, but the alien panel does not include the new pair, so the new-pair codes could not have been bent by it. Chapter 2 admits the pair and prints the envelope; Chapter 3 prints the two readings — the alien's in full, and the fourth blind coding of the widened ten-theory panel; Chapter 4 unblinds with the outgroup test as its headline; Chapter 5 convenes the court under the rules A4c locked before the alien ever ran; Chapter 6 renders the verdict and re-issues the kit at ten theories. The census-killing clauses stay armed throughout, exactly as they have stayed armed since the first failure census.

2. The Fifth Pair and the Envelope

The extension rule's four criteria, tried against the fifth pair. Dominance: caloric held the quantitative theory of heat from Lavoisier's installation — the ice-calorimeter memoirs with Laplace (1782-84), the element calorique at the head of the *Traité's* table of simple substances (1789) — through Carnot's engine analysis (1824) and Regnault's precision-thermometry program to the 1840s: sixty-plus years of dominance over heat physics, the new gas chemistry, and the engineering science of the steam engine, with Black's calorimetry upstream of it all. Death: no working physicist holds heat as a conserved, self-repelling fluid; the machinery — calorimetry, steam tables, Fourier's equations — survives in service, death of ontology and not of apparatus, the census's second criterion to the letter. The match is the strong one, and this pair carries it in a form no previous pair matched: three weapons forged inside the dead program where every earlier pair carried one.

The three weapons deserve their names, because they are the structural fact that makes this pair a natural experiment — and Table 1 prints them in the census's own form. First, Carnot's 1824 *Reflexions* was written entirely within the caloric frame — the falling-caloric water-wheel analogy, conservation assumed as the axiom — and its central result, the efficiency of any ideal engine bounded by the reservoir temperatures alone, was consumed by Thomson (1848-49: the absolute scale built on Carnot's function) and Clausius (1850: the re-derivation that discards conservation), the Tycho-archive parallel exact: the theorem assembled for the dead program's next refinement, used to delete its axiom. Second, Regnault's steam measurements (1847), commissioned inside the Paris program, became the data base of the 1850-51 synthesis. Third, the minted quantities themselves — Black's specific heats and latent heats, and the ratio Laplace used to correct the speed of sound (1816) — were converted by Mayer (1842) into the mechanical equivalent of heat: the dead program's own coins, spent by its executioner.

Dead theory (control)	Executioner (case base)	The discovery made inside the dead program
Phlogiston, 1703-1780s	Lavoisier's oxygen chemistry	Priestley's oxygen, 1774 — isolated by a loyalist, read as dephlogisticated air; the shared balance that weighed the calcined metals
The luminiferous ether, 1678-1905	Einstein's relativity	Michelson's interferometer, built 1887 to detect the medium — the null result arming the executioner
The epicycle tradition, 150-1609	Newton's mechanics (via Kepler)	Tycho's archive — the record assembled for the tradition's next refinement, consumed by Kepler to delete the circle
Miasma theory, c. 400 BCE-1880s	Germ theory (Pasteur, Koch)	Farr's register — the General Register Office archive built for the sanitary program's own questions; Snow's 1854 investigation ran on it, inside the Epidemiological Society the movement founded
Caloric, 1783-1850s	Thermodynamics (Joule, Clausius, Kelvin)	Three weapons: Carnot's 1824 theorem, written inside the caloric frame and consumed by Thomson and Clausius to delete its axiom; Regnault's 1847 steam tables, commissioned for the program; Black's minted quantities, converted by Mayer in 1842 into the mechanical equivalent

Table 1 — The census under its rule, extended to five matched pairs: each discovery that armed an executioner was made inside the dead theory's own program, on its own instruments, by its own defenders — the fifth pair carrying three weapons where every earlier pair carried one.

Thermodynamics enters by Volume II's method, and its scope line runs from Joule's mechanical-equivalent experiments (1843-49) and the Clausius-Thomson syntheses (1850-51) through entropy (1865) and the statistical-mechanical consolidation (Maxwell, Boltzmann) to about 1900. One honesty caveat is printed with the pair, because the record demands it: the caloric execution was staged, not instant — Joule's measurements were resisted for years, Kelvin's conversion took from 1847 to 1851, and Carnot's own posthumous notes (published 1878) show he had privately abandoned conservation before his death in 1832. And the pair carries one fact unique in the census, which 'print whatever differs' exists to catch: the dead theory's central theorem survived the death unchanged — Carnot's bound, derived under caloric conservation, is today's second law. No previous pair shows structure outliving ontology so completely, and the census's coding had never before had to price a corpse this strong.

Panel (alphabetical, as presented)	The scope line the coder received
T1 • The caloric theory of heat	Heat as a conserved, self-repelling fluid — Lavoisier's element calorique (Traité, 1789), with free and latent caloric, from Black's calorimetry (c. 1760) through Carnot's engine analysis (1824) and the Paris precision-thermometry program (Regnault) to the 1850s.
T2 • The epicycle tradition (Ptolemaic astronomy)	The deferent-epicycle-equant mathematical astronomy from Ptolemy's Almagest (2nd century CE) through the Islamic and European medieval traditions to Tycho Brahe's era (roughly 150-1600).
T3 • Germ theory of disease	The doctrine that specific microorganisms are the specific causes of specific diseases — from Pasteur's fermentation and spontaneous-generation work (1857-64) through Koch's anthrax, tuberculosis, and cholera bacteriology (1876-84) to the consolidation of medical bacteriology by 1900.
T4 • The luminiferous ether	The medium of light and electromagnetism, from Huygens's wave theory (1678) and Fresnel's optics through the Maxwell-Lorentz programs to 1905.
T5 • Miasma theory of disease	The doctrine that epidemic and endemic disease arises from foul air — miasmas given off by decaying organic matter, filth, and swamp exhalations — from the Hippocratic airs-waters-places tradition through the nineteenth-century sanitary movement to the 1880s.
T6 • Newtonian mechanics	The Principia's system of the world — the laws of motion plus universal gravitation (1687, editions 1713 and 1726).
T7 • Phlogiston chemistry	Stahl's theory from 1703 — combustion and calcination as release of phlogiston, metals as calx plus phlogiston — the framework of eighteenth-century pneumatic chemistry into the 1780s.
T8 • Quantum mechanics	From Planck's quantum hypothesis (1900) through the matrix and wave mechanics of 1925-27 and the Copenhagen consolidation.
T9 • Relativity	Einstein's special relativity (1905) and general relativity (1915), including their consolidation era.
T10 • Thermodynamics	The laws of heat and work — Joule's mechanical-equivalent experiments (1843-49), the Clausius and Thomson syntheses (1850-51), entropy (1865), through the statistical-mechanical consolidation (Maxwell, Boltzmann) to about 1900.

Table 2 — The widened panel of ten as the fourth coder received it: alphabetical, no verdict attached, the same instrument and evidence rule as every prior reading. The two new entries extend the census's own matched pairs; the alien, for scale, received the eight-theory panel the kit printed.

The envelope, drawn next, with everything inside it fixed before any coder ran. The instrument for the author's pre-registration: the post-A4c amended one — the four-plus-one discriminators, the core-posit deletion with its birth-window stipulation, the direction-split embedding, the graduated founders' scale, the as-written/operationalized dual for minted measurables. The instrument for the fourth blind reader: the original Volume I wording verbatim, exemplars intact, exactly as every blind coder before it received — so its convergence with (or divergence from) the amendments remains a finding rather than an input.

The panel: ten theories, alphabetical, one neutral scope line each, printed in Table 2 exactly as received. The forbidden list: no files, no web, no lookups; the point notes requested; the runtime model logged and printed — glm-5.3, the one lineage this environment supplies, printed as A4c printed it, with the difference that this time the alien beside it is not a slot but a file.

3. The Two Readings

Table 3 prints the alien reading without commentary — seventy-two cells, each returned with its named-evidence line, the complete output archived verbatim with this amendment. Three of its facts are visible before any comparison is attempted. The dead side reads lower than any family run has read it: the alien gives the ether and the epicycles partials on rows where every family reading fails them, and prices phlogiston's structure more generously than the census ever has. The living side breaks exactly once — Newton a partial on formalism and meaning, the same cell the third reading broke, replicated now from outside the family for the first time. And the germ-theory column is read strictly: derivation a partial, formalism a partial, founders failed — the physics-shaped points again, from a lineage that cannot be accused of sharing the family's physics training.

Point	Epicycles	Germ theory	Ether	Miasma	Newton	Phlogiston	Quantum	Relativity
P1 • Deletion at birth	P	S	P	P	S	P	S	S
P2 • Unification	P	P	S	F	S	S	S	S
P3 • Universal constant	F	F	P	F	S	F	S	S
P4 • Derivation-first	P	P	S	F	S	P	S	S
P5 • Conservative embedding	P	P	P	F	S	P	S	S
P6 • Formalism / meaning	P	P	P	F	P	P	S	S
P7 • Founders' resistance	P	F	P	F	S	F	S	S
P8 • Generative slope	S then D	S	S then D	S then D	S	S then D	S	S
P9 • Crisis-chaining	P	P	P	F	S	P	S	S

Table 3 — The alien reading, printed as delivered: a coder of a different model family, the kit's verbatim prompt, the eight-theory panel. The dead side reads lower than any family run; the living side breaks once — Newton P6, the elasticity the third reading measured, replicated cross-family.

Table 4 prints the fourth blind coding — ninety cells, the ten-theory panel's first reading, a fresh context of the author's family, blind, the original instrument verbatim. Its headline is the new pair. Caloric posts the strongest dead-side profile any reading has ever given a corpse — partials on structure, machinery, and measurables, the minted-measurables partial being a cell no dead theory had ever taken cleanly — and it matches the pre-registration cell for cell, a fact Chapter 4 prices for what it is. Thermodynamics posts six satisfied, two partial, zero failed, with its constant a partial and its founders a partial, matching the pre-registration on eight of nine — the one disagreement sitting exactly on the derivation-first cell the pre-registration itself flagged as the contestable one. The physics trio holds at twenty-six of twenty-seven, and the break is Newton's formalism cell again: two family readings and the alien, now, all breaking at the same joint.

Point	Caloric	Epicycles	Germ	Ether	Miasma	Newton	Phlogiston	Quantum	Relativity	Thermodynamics
P1 • Deletion at birth	F	F	S	F	F	S	F	S	S	S
P2 • Unification	P	P	S	P	F	S	P	S	S	S
P3 • Universal constant	F	F	F	F	F	S	F	S	S	P
P4 • Derivation-first	P	F	P	P	F	S	F	S	S	P
P5 • Conservative embedding	P	P	P	P	P	S	F	S	S	S
P6 • Formalism / meaning	P	P	F	P	F	P	F	S	S	S
P7 • Founders' resistance	F	P	F	P	F	S	F	S	S	P
P8 • Generative slope	S then D	S then D	S	S then D	S then D	S	S then D	S	S	S
P9 • Crisis-chaining	P	P	S	S	F	S	F	S	S	S

Table 4 — The fourth blind coding: a fresh context of the author's family, blind, the original Volume I instrument, the ten-theory panel. Caloric posts the strongest dead-side profile in the census's history; the physics trio breaks once, at Newton P6 again.

The two readings' point notes are the instrument audit this exercise never commissions and cannot dismiss — and for the first time they arrive from both sides of the lineage fence. The alien's eight flags include the timing of deletion at birth (spread over decades, not christened in a day); the

postulated-substance problem on unification — a unity built on positing a substance is not the revelation that a distinction was an artifact, stated with the ether as the sharpening case: the unification real, the unifier superfluous; the constant's ownership question (first measured, or newly universal?); the attribution rule when the successor keeps the mathematics and deletes the ontology — the alien proposing outright the rule the court had already adopted as doctrine; the founders' coin threshold (belief paid at founding, not costs imposed on the theory or paid by forerunners against incumbents); harvest attribution; and the deeper-floor versus deferred-question distinction on crisis-chaining, the epicycles its named case. Five of the eight match flags the family's readings had raised from the other side; each is handled in Chapter 5, under the same evidence rule as every coder's.

4. The Unblinding: The Outgroup Test

The outgroup test is the question this amendment was ordered to answer, and its headline number comes first. The family now has four blind runs — the two A4b coders, the third reading, the fourth — and their mutual weighted agreement spans 88.0 to 94.4 percent (mean 90.2, the two original coders' 94.4 at the top). The alien's agreement with those same four runs spans 86.1 to 87.5 percent, mean 86.9. The gap is 3.3 points: the outgroup sits below the family band, measurably, consistently — and with zero outright inversions against any family reading, every single disagreement landing one step away, adjacent, never opposed. The verdict the census prints is therefore double-edged and exact. The lineage effect is real: readings of the same instrument differ by family at a distance the family's own runs do not match, and the genetic-dependence confession was load-bearing, not ritual. And the effect is small and monotone: a calibration offset, not a worldview — no cell anywhere in the record flips from satisfied to failed across the lineage fence.

Where the distance lives is the finding under the headline, because it is not scattered. Against the two A4b coders, the alien's per-point agreement bottoms at formalism and meaning — 66.7 percent, the floor of the entire table — with conservative embedding at 79.2, deletion at 83.3, and derivation-first at 87.5; the constant, the slope, and crisis-chaining sit above 91. Against the third and fourth readings, the soft band repeats: deletion 78.1, derivation-first 78.1, unification 81.2. These are, cell for cell, the instrument's declared elastic joints — the presence-versus-profundity ambiguity on the formalism point, the birth-window and direction-split stipulations, the two-births problem — every one of them printed in the census's own amendment record before the alien ever ran. The census's amendment map predicts where an outgroup will part from it. The disagreements are not noise in the reading; they are the instrument's own labeled soft spots, seen from the other side of the lineage fence.

The alien against the census's own history tells the same story from a different angle, and its two cleanest facts belong together. Against the census as originally written, the alien agrees on the old dead side at 72.2 percent with two outright inversions — and both of them, inspected, land on cells the census had already revised before the alien ran: the ether's derivation-first, where the alien's satisfied sits one step beyond the amended partial the A4b court adopted; and phlogiston's founders, where the alien's failed is the A4b refutation itself, arrived at independently, from outside the family, without ever having seen the

amendment. Against the amended dead side of four theories the alien reads 79.2 percent; against the living trio, twenty-six of twenty-seven, its single break the Newton formalism cell. The alien, in short, lands on the amendments rather than on the originals the author first wrote — the substitution's question from A4c, answered now from outside the school.

Comparison	Cells	Weighted	Exact / Adj / Dist
Alien (gpt) vs A4b coder A	54	87.0%	40 / 14 / 0
Alien (gpt) vs A4b coder B	54	87.0%	40 / 14 / 0
Alien (gpt) vs third reading X3	72	86.1%	52 / 20 / 0
Alien (gpt) vs fourth reading B4	72	87.5%	54 / 18 / 0
B4 vs A4b coder A	54	88.0%	41 / 13 / 0
B4 vs A4b coder B	54	89.8%	43 / 11 / 0
B4 vs third reading X3	72	91.7%	60 / 12 / 0
X3 vs A4b coders A and B (A4c, for scale)	54	90.7%	44 / 10 / 0
A4b coder A vs A4b coder B (A4b, for scale)	54	94.4%	48 / 6 / 0
Alien (gpt) vs census v1 (old dead side)	27	72.2%	14 / 11 / 2
Alien (gpt) vs amended census (dead side)	36	79.2%	21 / 15 / 0
B4 vs amended census (dead side)	36	86.1%	26 / 10 / 0
B4 vs the A4d pre-registration (new pair)	18	97.2%	17 / 1 / 0

Table 5 — The unblinding, all readings: the family's four runs span 88.0-94.4 percent mutual agreement; the alien spans 86.1-87.5 against them — 3.3 points below the band, zero inversions. The new pair's pre-registered coding holds at 97.2 under the fourth blind reading.

The fourth reading's numbers price the new pair and the family's stability. Against the census as originally written, 77.8 percent with a single inversion — phlogiston's founders again, landing on the already-revised cell. Against the amended dead side of four theories, 86.1. Against the amended germ profile, 94.4. The living trio, twenty-six of twenty-seven, the break Newton's formalism cell — the quantum's derivation cell, broken once by the third reading, does not break again, and is recorded as the rarer elastic joint. And against the pre-registration locked before it ran: 97.2 percent over the new pair's eighteen cells — caloric exact, cell for cell, all nine; thermodynamics eight of nine, the disagreement sitting on the derivation cell the pre-registration itself had named contestable. Table 5 assembles the numbers; Table 6 the per-point bands; the court's docket follows from them.

Signature point	Alien vs A4b A+B	Alien vs X3+B4	The disagreement's character
Deletion at birth	83.3%	78.1%	The birth-window: deletions spread over decades, and the christening question the census had to stipulate.
Unification	95.8%	81.2%	Postulated-substance unity versus dissolved joints — the alien's own flag, matching the demotion.
Universal constant	95.8%	93.8%	The sharpest row holds cross-family — with the ether's ownership dissent the one recorded break.
Derivation-first	87.5%	78.1%	The two-births problem and the founding's instruments.
Conservative embedding	79.2%	87.5%	The direction-split and the attribution rule — the softest joint, from both sides of the fence.
Formalism / meaning	66.7%	84.4%	The floor of the table: presence versus profundity, flagged by every reading that ever broke the cell.
Founders' resistance	91.7%	93.8%	The coin threshold: whose surrender counts, paid when.
Generative slope	91.7%	100.0%	Every dead tradition generative-then-degenerating, no living theory — unanimous with the fourth reading.
Crisis-chaining	91.7%	84.4%	Deeper floor versus deferred question versus fatal anomaly.

Table 6 — Per-point weighted agreement for the alien reading: against the two A4b coders (twelve comparisons per point) and against the third and fourth readings (sixteen cells per point). The floor is the formalism point; the soft band is the census's own amendment record.

The docket, before the court convenes. Three cells now carry post-amendment reading records that no longer support the census's cells, and in each the alien's vote is among the movers. Newton's formalism cell: the two A4b coders satisfied, and then the third reading, the fourth, and the alien all partial — three consecutive post-amendment readings plus the outgroup against the census's satisfied. The ether's embedding cell: the two A4b coders satisfied, the third and fourth readings partial, the alien partial — the author's failed, defended since A4b by the successor-recovery refinement, now the minority reading in the record. And miasma's crisis cell: three post-amendment readings — third, fourth, and the alien — all failed against the census's partial. The motion rule A4c locked before the alien ran binds the author in advance: where the outsider's reading and the blind readings agree against the census, the census moves. Chapter 5 tries the docket under it.

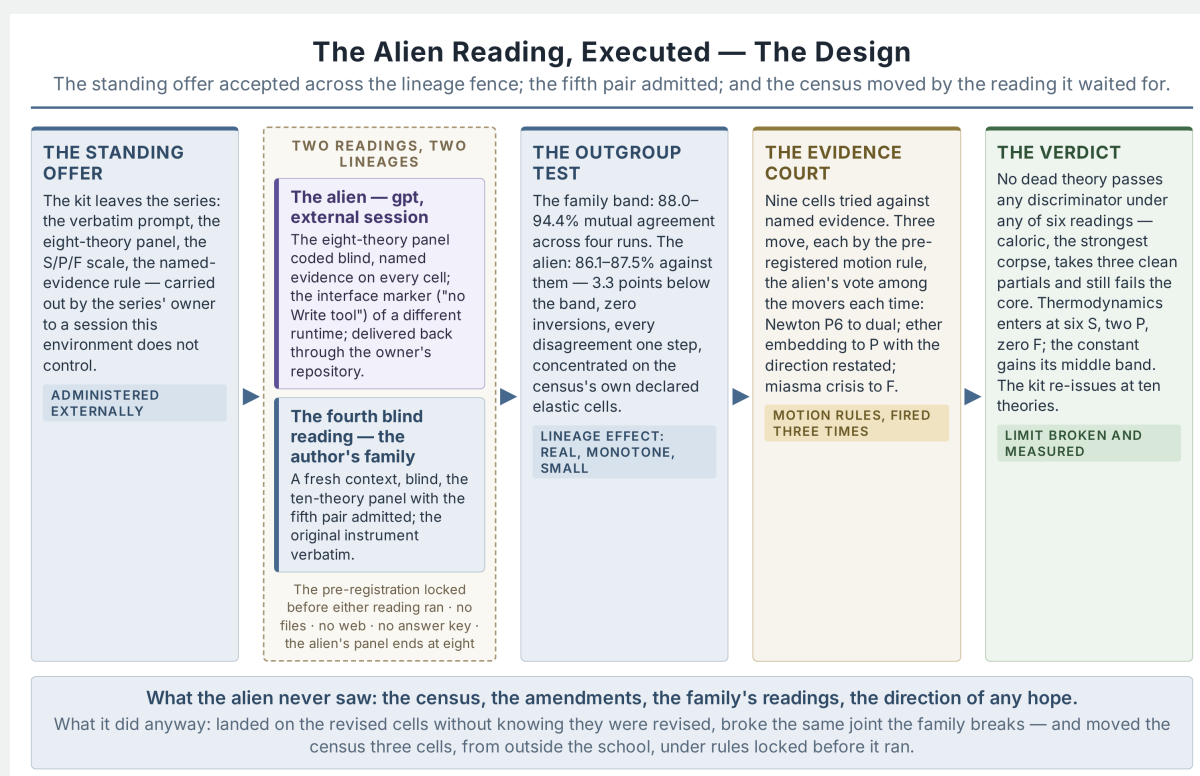


Figure 1 — The A4d design: the replication kit leaves the series through its owner, is administered to a reader of another family (gpt), and returns as the alien reading; the fourth blind coding widens the panel to ten; the outgroup test measures the lineage gap (3.3 points, zero inversions); the court moves three cells under the pre-registered motion rules; the kit re-issues at ten theories.

5. The Adjudication

The court reconvenes under the rules A4c locked before the alien ever ran, and they are restated before any ruling: evidence over authorship, every contested cell tried against named historical fact; the motion rule — where the outsider's reading and the blind readings agree against the census, the census moves; and the census-killing clauses stay armed. The docket has three parts — the motion cells, the contested cells of the new pair, and the defended dissents — and Table 7 is its ledger. What makes this session different from every court before it is written into each movement: the alien's vote is among the movers, and the movements are therefore the reading's material effect on the census, not its decoration.

Newton's formalism cell, tried first because the record is now unambiguous. The readings: the two A4b coders satisfied; the third reading, the fourth, and the alien all partial — three post-amendment family readings plus the outgroup, against the census's satisfied. The evidence both ways: on one side, the inverse-square law was common currency before the Principia — Halley, Wren, and Hooke all held it from Kepler's third law plus Huygens's centrifugal measure, and Halley's 1684 visit posed the orbit question Newton's dynamics answered — so the profound move was new mathematics plus new system, not an interpretation of inherited mathematics; on the other, the formalism was accepted for a century and a half while its meaning was fought over, hypotheses non fingo in 1713 to Mach's re-reading in 1883, which is dissociation exactly as the point writes it. The ruling: the cell moves to the dual form — satisfied as written, partial on the interpretation-arrival reading — the living side's first dual cell, and the

presence-versus-profundity flag that A4c adopted is now a cross-family fact of the instrument rather than one school's scruple.

The ether's embedding cell, tried second, because the pre-registration's own convention is at stake. The readings: the two A4b coders satisfied, the third and fourth readings partial, the alien partial — the author's failed, defended since A4b by the successor-recovery direction, now the minority position in the record. The evidence: Maxwell's equations survived the medium's deletion without a changed symbol, and Fresnel's dragging coefficient — conjectured from Arago's 1810 null, confirmed by Fizeau in 1851 — is recovered exactly as the first-order term of the relativistic velocity addition, a theorem surviving as a limit; the medium itself, meanwhile, received no domain of validity anywhere. The ruling: the composite cell moves to partial, and the discriminating face is restated as the direction the census always meant — successor-recovery of the ontology, on which every dead theory in the panel fails without exception. With the structure-versus-craft distinction now doctrine — theorems survived the ether, the epicycles, and caloric; craft survived miasma; nothing survived phlogiston — the composite records what survived, and the direction keeps the discrimination. The demotion is a restatement, not an inflation: the fifth pair forced it by being the first quantitative corpse.

Miasma's crisis cell, tried third. The readings: the census's partial — pre-registered in A4c and sustained there against the third reading's dissent — now against the third reading, the fourth, and the alien, all failed. The evidence, both ways: the sanitary program's archive and its environmental question armed Snow and Koch, which was the A4c ruling's ground; but the doctrine itself closed no crisis — it was the inherited frame, not a founding — and it opened no deeper floor: it faded, its last theoretical gesture (Pettenkofer's groundwater variant, 1869-73) generating no successor program. The archive's productivity is real, and it belongs to the apparatus — the registers, the infrastructure — which CR-2 already credits separately as death-of-ontology-not-of-apparatus. The ruling: the cell moves to failed, the census against its own pre-registered coding, the motion rule executed a third time; the oldest corpse in the history of science is now also the emptiest profile the census has ever recorded — one partial in nine cells.

The contested cells of the new pair, tried under the same rules, with the fourth reading as the blind evidence and the alien silent (its panel ends at eight). Caloric's profile stands as pre-registered — deletion failed, the substance inherited and renamed by the *Traité* while the famous deletions in the same book belong to the oxygen pair's christening; unification partial, vocabulary-level; the constant failed; derivation partial, the structure-not-content rule's strongest case, Carnot's theorem surviving the frame that derived it while the friction anomalies were retrofitted; embedding partial, the theorems surviving verbatim with no parameter limit in sight; measurables partial — minted at a strength no dead theory had shown, the killing coin the successor's; founders failed, Carnot's private recantation never tendered; the slope generative-then-degenerating; crisis partial, the motive-power question opened inside the program becoming the successor's foundation. Thermodynamics, with its two contested cells: derivation-first stands satisfied — the first law is the textbook case of multiple independent discovery, Mayer's 1842 derivation from the old gas data and Helmholtz's 1847 synthesis needing no new instruments, with Joule's

measurements the third road — and the two-roads stipulation (the derivation road and the experimental road, both founding the same law) enters the instrument beside the birth-window; the dissent is printed with its evidence. The constant stands partial — the J/k dual structure, printed in full in the pre-registration — and with it the domain-marked discriminator gains its first within-physics middle band: caloric failed, thermodynamics partial, the trio satisfied.

The defended dissents, printed with their evidence and left standing. Germ theory's founders cell holds at partial with the record split — the third reading partial, the fourth and the alien failed — because the evidence is specific: Pasteur's surrender of the chemical-catalysis theory of fermentation he absorbed from his own formation, paid publicly against Liebig to the elder's death in 1873, is payment; the dissenting readings' evidence — costs paid by forerunners against incumbents, Semmelweis's ruin above all — is real and prices other people's coins, not the founders' own; the cell is recorded as elastic, the graduated scale holding at compromise-to-surrender. The ether's constant cell holds failed under the ownership rule — a constant is credited to the frame that makes it invariant, not the frame that measures it — with the alien's partial recorded as the ownership question's cross-family elasticity, its own point note naming the same fork. Miasma's embedding cell holds failed — the craft that survived was engineering, not theory, and the structure-versus-craft distinction is now the doctrine's answer to the fourth reading's dissent. And the two as-written formalism cells — the ether's and the epicycles' — hold their duals, their cross-family splits recorded in the ledger as the presence-versus-profundity joint's measured width.

Contested cell	Readings	The evidence the court weighed	Ruling
Newton • formalism	S / P,P,P	The inverse-square law common currency before 1684; the formalism fought over from 1713 to Mach — both readings of the point in one cell.	Dual: S as written / P on the arrival reading — the motion rule fired
Ether • embedding	F / S,S,P,P,P	Maxwell's equations and Fresnel's coefficient survive (the latter as the first-order limit); the medium recovered as no limit anywhere.	Revised to P; successor-recovery of ontology restated as the discriminating face
Miasma • crisis	P / F,F,F	The archive's productivity belongs to the apparatus (CR-2); the doctrine closed no crisis and faded, Pettenkofer's variant generating no successor.	Revised to F — the census moves against its own pre-registration
Germ • founders	P / P,P,F,F	Pasteur's Liebig surrender, paid publicly 1857-73 — against costs paid by forerunners, which price other people's coins.	P stands; elasticity recorded
Ether • constant	F / F,F,F,P	Maxwell's c-ratio measured inside the program; the invariant speed installed by 1905 — the ownership rule.	F stands; the ownership fork now cross-family

Contested cell	Readings	The evidence the court weighed	Ruling
Miasma · embedding	F / F,F,P	The surviving sanitation was craft, not theory-structure; no theorems existed to survive.	F stands; structure-versus-craft adopted
Caloric · profile	prereg / B4	Nine cells pre-registered before the fourth reading ran; the blind reading returns the identical nine — the strongest corpse: partials on structure, machinery, measurables.	Stands as pre-registered, cell for cell
Thermo · derivation	S / P	Mayer 1842 from the old gas data; Helmholtz 1847; Joule's measurements the third road — the simultaneous-discovery record is the over-determination itself.	S stands; the two-roads stipulation adopted; dissent printed
Thermo · constant	P / P	J minted at the founding, a unit bridge demoted to a definition; k minted by the consolidation — the J/k dual.	P stands — the constant's middle band, in physics

Table 7 — The adjudication ledger: nine cells tried, three revised — every revision by the pre-registered motion rule, the alien's vote among the movers each time — one dual created, five sustained, the new pair's profile standing as pre-registered. The author is overruled on his own cells three times, which is what pre-registering is for.

6. The Verdict: The Limit Broken and Measured

The registration, clause by clause. The census survives its sixth reading, its first from another family: no dead theory — three physics corpses, the oldest framework in the history of science, and now the strongest quantitative corpse — passes any of the five discriminators under any reading ever run, and the ether's lone as-written satisfied remains the adjudicated confound it has been since A4b. Three cells moved, every one by the motion rule locked before the alien ran, the alien's vote among the movers each time: Newton's formalism cell to the dual form, the ether's embedding cell to partial with the discriminating direction restated, miasma's crisis cell to failed. The living side admits its strongest second resident — thermodynamics, six satisfied, two partial, zero failed — and pays for it with the constant's middle band: caloric failed, thermodynamics partial, the trio satisfied, the domain-marked point now printing a gradient inside its own domain instead of pretending to a clean line.

Signature point	Phlogiston	The ether	Epicycles	Miasma	Caloric
Deletion at birth	F — Becher's principle renamed; nothing deleted	P * — the wave program deleted the corpuscle; its own center an inherited medium reinstated	F — extended Eudoxus; circularity never touched	F — the consolidated form deleted nothing	F — the fire-principle renamed and quantified; the Traité's deletions were the oxygen pair's christening
Unification	P * — a posited substance is not a dissolved joint	S — one medium under optics and electromagnetism	P * — one toolkit across the planets; vocabulary, not law	P — the zymotic class: administrative, not natural	P — one substance's bookkeeping; the substance-versus-motion division begged
Universal constant	F — no quantitative core; weight forced phlogiston negative	F — c belonged to the fields and outlived the medium	F — dozens of parameters, zero invariants	F — nothing constant; fitted coefficients per epidemic	F — material properties; Dulong-Petit the exception-ridden regularity the successor explained
Derivation-first	F — each new gas retrofitted, after the fact	P * — real derivational firsts; the defining note dishonored in 1887	P — parameters from centuries of records: content, never structure	F — the incumbent; data broke it rather than being derived	P — Carnot's theorem survived the frame; the friction anomalies retrofitted
Conservative embedding	F — deleted without remainder by oxygen chemistry	P * — the equations survived (Fresnel's coefficient as the first-order limit); the medium recovered as no limit — direction: F	P — devices survive as Fourier; the ontology deleted	F — the surviving sanitation was craft; no theorems existed to survive	P — the theorems survive verbatim (Fourier, Carnot); no parameter limit
Formalism / meaning	F — no formalism; no measurable ever minted	S as written * — discriminates only operationalized (A6)	S as written * — the operationalized form discriminates	F — no symbolic surface; no unit of miasma ever named	P — minted (Black's latent heat, the calorie, the calorimeter); the killing coin was the successor's

Signature point	Phlogiston	The ether	Epicycles	Miasma	Caloric
Founders' resistance	F † — Priestley defended, paid nothing: defense is not payment	P * — Lorentz surrendered detectability, kept the medium	P * — Ptolemy's equant: a founder's compromise, collected for fourteen centuries	F — refusal at the price of the body itself	F — Carnot's private recantation, published 1878, was never tendered
Generative slope	S then D — the gas family; then negative weight	S then D — Fresnel to the interferometer; then the incoherence triangle	S then D — fourteen centuries; then the equant wound	S then D — the sanitary acts and the register; then the restatements	S then D — calorimetry to Carnot; then the capacity epicycles against Joule
Crisis-chaining	P — opened the air questions that killed it	S as written * — closed the corpuscular crisis, opened the drift problem; demoted regardless	P — the equant objection opened its execution	F ‡ — it faded; the archive was apparatus, not doctrine (revised this run)	P — the motive-power question, opened inside the program, became the successor's foundation

Table 8 — The census matrix after the fifth pair and the alien reading: five dead theories, the dead side now a graded slope from the emptiest profile ever recorded to the strongest corpse. Cells marked * revised this run or an earlier amendment's; † refuted outright; ‡ revised by the motion rule this run. The five-discriminator core holds on every column, on the restated faces.

The genetic-dependence limit's new status, stated at the weight the series owes its loudest confession. Broken, as an availability fact: a cross-family reading exists, archived whole, and it did not merely validate — it moved the census, three cells, its vote the decisive one where the family's post-amendment readings were already leaning. Measured, for the first time: 3.3 points below the family band, zero inversions, the disagreements concentrated cell for cell on the instrument's declared elastic joints. And therefore reduced to what it always really was: a calibration question, not a validity question. The amendments replicate across the lineage precisely where they are amendments of the instrument — the alien lands on the ether's and phlogiston's revised cells without ever having seen them; the school's readings are not the world's; and the difference between them is now a number with a location, printed above, where any future alien reading can be measured against it. The day A4b promised has arrived, and the census is smaller, harder, and — for the first time — partly not the author's school's.

Signature point	Germ theory (living)	Thermodynamics (living, new)
Deletion at birth	S — the double deletion: spontaneous generation and foul-air etiology	S — the double deletion: the substance of heat and unlimited convertibility
Unification	S — the brewer's vat and the surgical ward, one cause-class	S — heat and work one currency; the firebox and the falling weight the same phenomenon
Universal constant	F — domain-marked: no constant to install; the point abstains outside physics	P — the J/k dual: a conversion constant demoted to a definition; the true invariant minted by the consolidation — the middle band, in physics
Derivation-first	P * — promissory notes honored; the founding carried by its own new experiments	S — over-determined by three roads (Mayer's derivation, Helmholtz's synthesis, Joule's measurements); the simultaneous-discovery record is the over-determination itself
Conservative embedding	P — the practice recovered, the ontology deleted; no limit in which foul air causes anything	S — the strongest embedding in the census: Carnot verbatim, Fourier untouched, the no-work sector recovered with the reason why
Formalism / meaning	F as written / S operationalized — the organism minted: stainable, countable, cultivable	S — absolute temperature, the joule, entropy minted; the efficiency bound a discriminating list that returns negatives
Founders' resistance	P — Pasteur paid Liebig his chemistry; Koch's tuberculin a cost imposed — elastic: two readings dissent	P — Kelvin's tendered coin: formed in the caloric physics, publicly surrendered 1849-51, his own scale's foundation rebuilt — compromise-to-surrender
Generative slope	S — virology, immunology, antibiotics, molecular genetics: still generating	S — Gibbs, the third law, the quantum, Shannon's entropy, black holes: the deepest cross-domain harvest on record
Crisis-chaining	S — closed the etiological deadlock; opened the host, the filterables, the carrier state	S — closed the motive-power question; opened time's arrow, 175 years and counting

Table 9 — The living residents after the fifth pair: germ theory's profile unchanged from A4c; thermodynamics the strongest second resident the case base has admitted — and the first to give the domain-marked constant a middle band.

The amended signature, stated whole, and printed in Tables 8 and 9 against the five dead theories and the two living residents. Four domain-general discriminators: deletion at birth, with the core-posit and birth-window refinements; successor-recovery of the ontology — the embedding point's restated face, on which every dead theory fails; generative slope, the trend; minted measurables, operationalized. One domain-marked discriminator with a middle band: the universal constant, physics-relative and gradient-printing. Two restated points, now doubly restated: derivation-first as structure-not-content with the birth-window and two-roads stipulations; founders' resistance as payment-not-defense, graduated, with the refused-coin test. Two demoted: unification, its dead-side leg substance-narrowed;

crisis-chaining, its weakest dead-side cell now weaker still. And the record now carries a gradient where it once carried a wall: the dead side runs from miasma — one partial in nine, the emptiest profile ever recorded — through phlogiston, deleted without remainder, to the structure-survivors, the epicycles, the ether, and caloric, the strongest corpse, three clean partials deep — while the living side runs from germ theory to thermodynamics to the trio, and the first dual cell sits on Newton. The census no longer sorts its panel into two heaps; it grades a slope.

The kit converts the clause from a purchase the environment cannot make into a standing offer the world can accept.

The Widened Census, Chapter 6 — the offer was accepted; the receipt is Chapters 3 through 5 of this document; the offer re-issues at ten theories.

The limits, at the weight the audit taught, and the kit renewed. One alien reading is a first measurement, not a law: the outgroup number is $n=1$, its family attested by delivery and interface markers rather than verifiable from inside, and a second family's reading could sit elsewhere in the band the first one defined. The adjudicator was again the author — mitigated, as before, by pre-registrations locked before runs and evidence printed for every ruling — and overruled three times, which is the design working, not the design failing. The count remains small and fame-confounded, and the fifth pair is the most famous pair heat owns. What this document does not claim is the larger arc, unchanged from the first census: a coding is not the world; the domain marks and the middle bands are confessions, not triumphs; and the world's ratification remains the only evidence this series acknowledges it lacks. The kit re-issues with this amendment at ten theories — the instrument, the panel, the scale, the evidence rule, the scoring scheme, and the adjudication rules verbatim, the widened prompt printed and archived — so that the next alien reading, from any family, run by anyone, anywhere, can test the ten and not only the eight. The ledger is open. Continue.

1 alien reading delivered across families the genetic-dependence limit broken as an availability fact — and its vote moved three cells	3.3 points the outgroup sits below the family band the lineage effect, measured: real, monotone, zero inversions	10 theories in the panel after the fifth pair no dead theory passes any discriminator under any of six readings
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The alien reading's own numbers: the offer accepted, the lineage gap measured, and the panel the next alien will be handed.